

ClipMind AI: Video Summarization & KeyMoments Detection

Milestone 3: Key Moments Detection and Analytics Dashboard (Module 5 and Module 6)

1. My Work Responsibility

- **Module 5 (Key Moments Detection):** Backend APIs for Transcript, Key Moments, Highlights and Keywords; API and schema validation; response-field consistency; error handling; authentication verification; API testing.
- **Module 6 (Analytics Dashboard):** Backend APIs for Usage Reports and Content Insights, built on top of the existing Analytics API; schema design; computation services; API testing against real data.
- **Milestone-wide:** Fixing the API and pipeline bugs that surfaced during real-video testing, and documenting the endpoints and their request and response shapes.

2. Project Objective

The objective of Milestone 3 was to take the key-moments and analytics foundation already built in the earlier milestones and make it a validated, real-data, end-to-end system. This milestone was scoped as integration, validation, fixing and demo-readiness rather than a rebuild: a real uploaded video flows through transcription, key moment detection, highlights, keywords, analytics, content insights and usage reports, with no hardcoded or sample content anywhere in the path.

My objective was to make the backend API layer for these features correct, consistent and reliable, to fix the defects that stopped the pipeline from producing real timing data, and to build the two analytics features that had no backend implementation at the start of the milestone: Usage Reports and Content Insights.

3. Project Description and Main Features

The complete processing flow for this milestone is:

Upload, FFmpeg audio extraction, Whisper transcription with timestamped segments, Key Moments, Highlights, Keywords, Analytics, Content Insights, Usage Reports, Display of results.

Module 5: Key Moments Detection

Module 5 detects the important segments (key moments) in a video's transcript, generates highlights from the top-ranked key moments, and extracts keywords. The results are stored inside the `Video` document in MongoDB.

Data models. Three embedded models were defined: `TranscriptSegment` (start time, end time, text), `KeyMoment` (start time, end time, label, text, importance from 0.0 to 1.0) and `Keyword` (word, normalized score from 0.0 to 1.0). The `Video` document holds `transcript`, `transcript_segments`, `key_moments`, `highlights` and `keywords`.

Backend APIs I worked on:

- **Transcript API** (`GET /api/videos/{video_id}/transcript`): returns the transcript and its timestamped segments once available; returns 404 if the transcript is not yet available.
- **Key Moments API** (`POST /api/videos/{video_id}/key-moments`): detects key moments from the video's real transcript segments and returns them with timestamps and importance scores.
- **Highlights API** (`POST /api/videos/{video_id}/highlights`): selects the top 3 key moments by importance score; requires key moments to exist first.
- **Keywords API** (`POST /api/videos/{video_id}/keywords`): extracts keywords using deterministic word-frequency analysis.

Keyword extraction method. The transcript is lowercased and tokenized, stopwords and short or non-alphabetic tokens are removed, words are ranked by frequency, and scores are normalized against the most frequent word (score = count / max count).

Error handling. Each endpoint returns clear errors: 400 when the required input is not yet available (transcript or key moments), 404 for an unknown video or missing transcript, 409 when generation is already in progress for the video, and 500 when generation fails (the failure is caught and logged to `error_message`).

Module 6: Analytics Dashboard (Usage Reports and Content Insights)

The core Analytics API (per-video analytics and the dashboard summary) was already built and verified. In this milestone I extended the Analytics Dashboard with two features that previously had no schema, no service logic and no API route. Both are computed on the fly from existing `Video` documents, following the pattern already used in the analytics service, so no new database collection was needed.

Usage Reports (`GET /api/analytics/usage-report`) shows upload activity and storage usage over time for the authenticated user's videos:

- Total uploads and uploads grouped by calendar day.
- Success, failed and in-progress counts, derived from the video status, and the overall success rate (guarded against division by zero).
- Total storage and average file size, read from the actual files on disk, with a safe fallback to zero if a file no longer exists.

Content Insights (`GET /api/analytics/content-insights`) shows content-level patterns across the authenticated user's completed videos:

- Top topics, counted by the number of different videos in which a keyword appears, so that broadly recurring themes surface instead of one video's noise.
- Average key moments per minute, using video duration estimated from the last transcript segment (guarded against division by zero).
- Average keywords per video.
- Length distribution: short (under 60 seconds), medium (60 to 300 seconds) and long (over 300 seconds).
- Only videos with status "done" are analyzed; incomplete videos are excluded.

4. Technology Used

- **Transcription and Audio:** OpenAI Whisper and FFmpeg
- **Analysis Methods:** Deterministic content-based scoring and word-frequency analysis (no new AI model introduced)

5. Screenshots

Insert the screenshots below from your Swagger UI or Postman runs before submitting.

Figure 1: Transcript API response (transcript text with real timestamped segments)

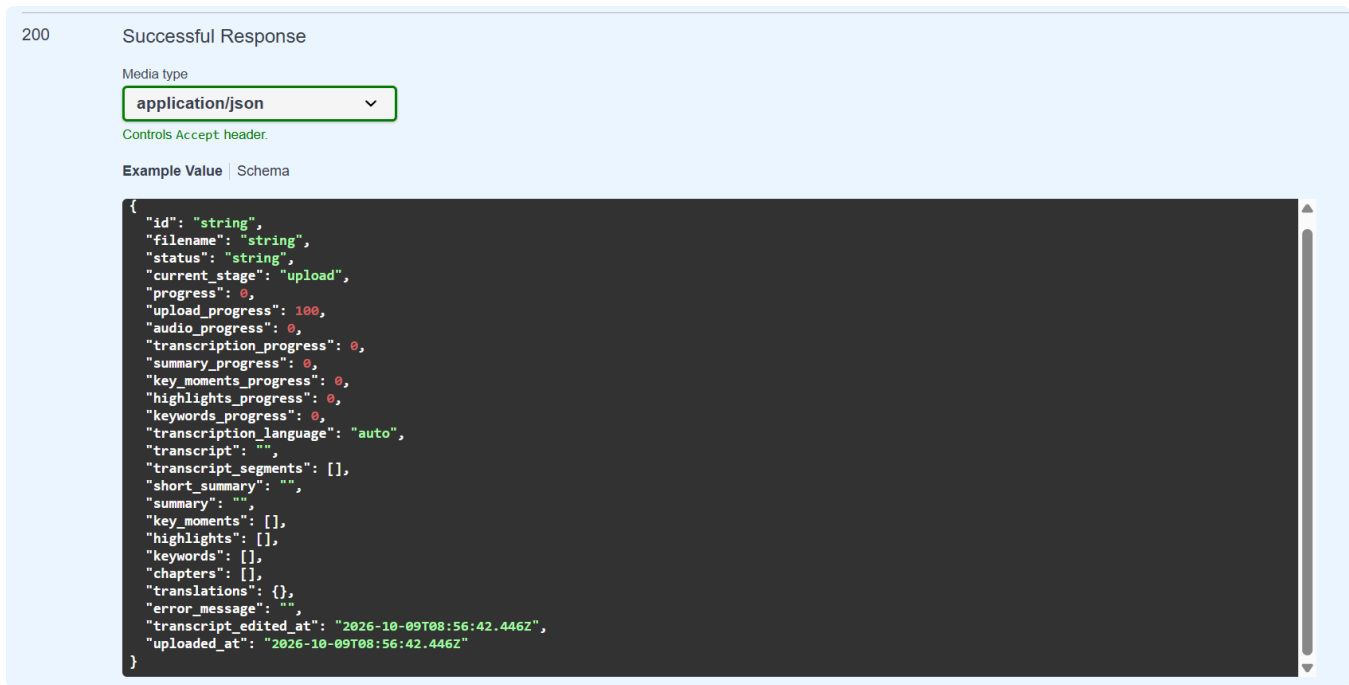
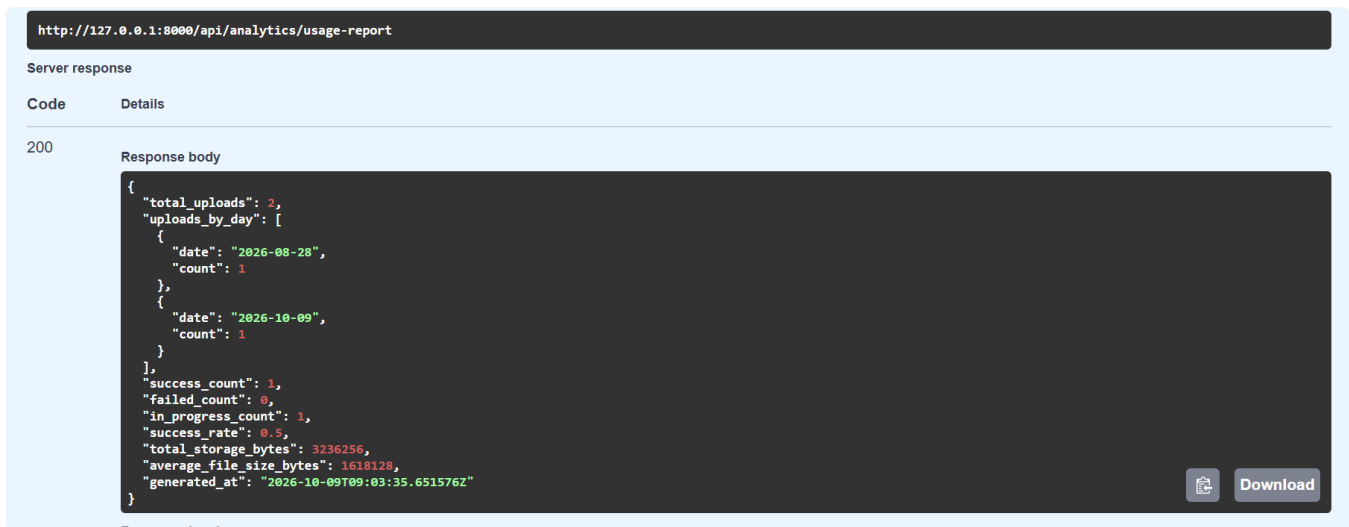


Figure 2: Usage Report API response (uploads by day, success and failure counts, storage)



6. Testing and Results

The APIs were tested live on real uploaded videos.

- **Module 5** (two real videos, 41 and 46 seconds): transcript segments, key moments, highlights and keywords all returned real timestamps and correct scores, and the analytics counts matched.
- **Error handling:** an unsupported format or empty file returned 400, a missing token returned 401, and an invalid video ID returned 404.

- **Usage Reports** (7 real videos): the success, failed and in-progress counts (4 + 3 + 0) matched the 7 uploads, and storage totals matched the file sizes on disk.
- **Content Insights** (4 completed videos): short videos were bucketed correctly and top topics aggregated across videos.

7. Challenges Faced

- **Real timestamps were being lost.** Whisper's segment timestamps were discarded, so key moments fell into fixed 30-second blocks. I changed transcription to return the segments and saved them to the video record.
- **Pipeline bugs.** I fixed the wrong argument order in the `detect_key_moments` call, the `start / end` versus `start_time / end_time` key mismatch, the wrong content in `models/video.py`, and the missing `app/services/__init__.py`.
- **Usage Reports and Content Insights had no definition.** I chose a scope based on what could be computed from existing data and built both on the same on-the-fly pattern as the existing analytics endpoints.
- **Open limitation.** Whisper segments have no gap between them, so the 1.5-second merge threshold collapses most videos into one or two key moments. I recommend a maximum moment duration or merging by sentence or topic boundary.

8. Conclusion

This milestone made the key moments and analytics features work on real data. On the backend, I fixed the defects that prevented real timestamps from reaching key moment detection, delivered the Transcript, Key Moments, Highlights and Keywords APIs with consistent responses and proper error handling, and built the Usage Reports and Content Insights APIs that extend the Analytics Dashboard. The APIs are secured with JWT, connected to MongoDB and tested on real videos, including invalid input, unsupported formats, empty files and authentication failures. One algorithm-tuning item, the merge behavior of key moments, remains open with a clear recommendation for follow-up.